

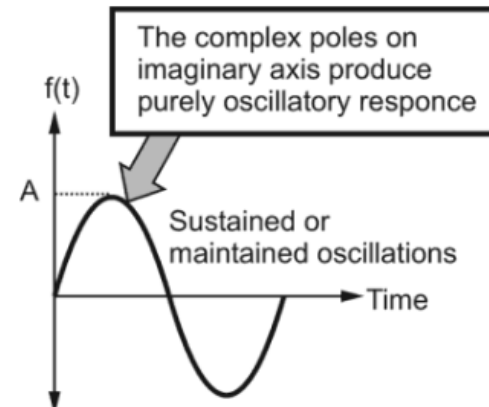
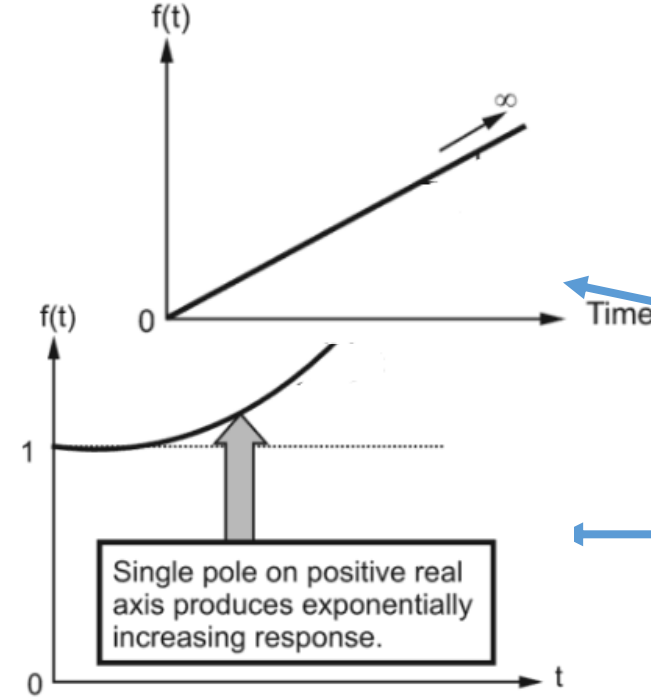
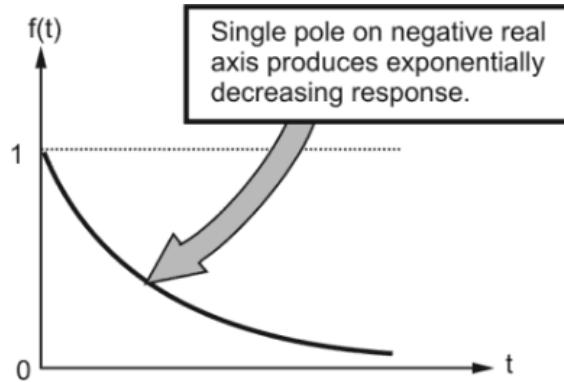
Outline

- Introduction to Network Functions
- **System Realizability and Stability**
- Derivation of Network Function from Partial Information

Process of Circuit Synthesis

- A network function is tested first for its realizability.
- After that, we proceed to realize the network
- Two important properties for realizability are
 - **Causality:** a voltage cannot appear at any port before a current is applied or vice-versa (i.e. the transfer function in time domain verifies $f(t) = 0$ for $t < 0$)
 - **Stability**

Recall: Examples of Laplace Transforms

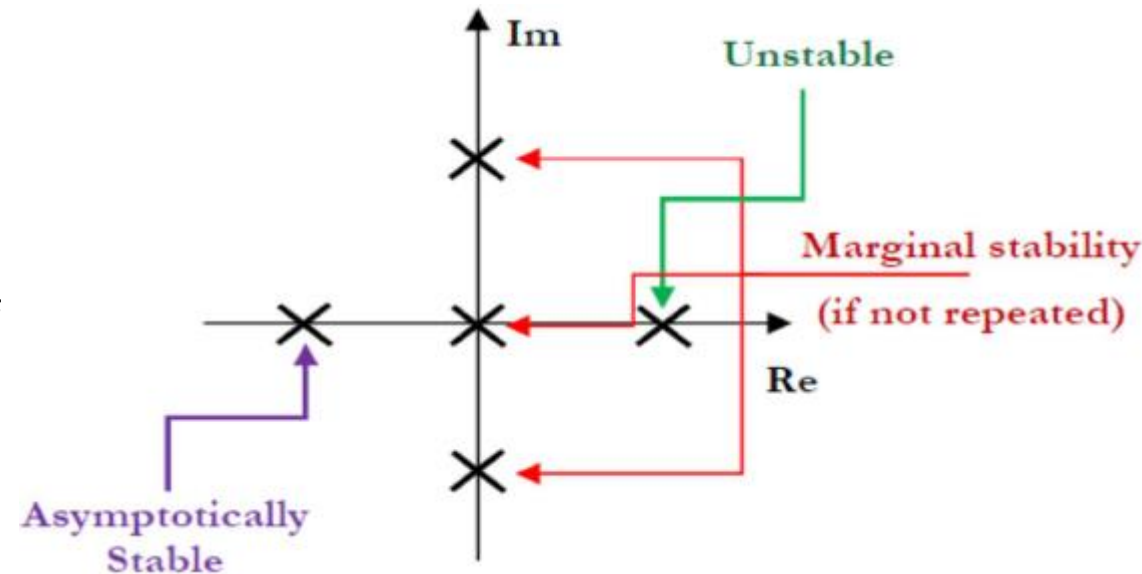


$f(t), t \geq 0$	$F(s)$
1. $\delta(t)$	1
2. $u(t)$	$\frac{1}{s}$
3. t	$\frac{1}{s^2}$
4. t^n	$\frac{n!}{s^{n+1}}$
5. e^{-at}	$\frac{1}{s+a}$
6. te^{-at}	$\frac{1}{(s+a)^2}$
7. $t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
8. $\sin bt$	$\frac{b}{s^2 + b^2}$
9. $\cos bt$	$\frac{s}{s^2 + b^2}$
10. $e^{-at} \sin bt$	$\frac{b}{(s+a)^2 + b^2}$
11. $e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$
12. $t \sin bt$	$\frac{2bs}{(s^2 + b^2)^2}$
13. $t \cos bt$	$\frac{s^2 - b^2}{(s^2 + b^2)^2}$

Stability of Network Functions

Necessary and sufficient conditions

- A system is stable as long as a bounded input produces a bounded output.
 - This type of system is known as bounded input bounded output (BIBO) stable system
- Necessary and sufficient conditions of a network function (i.e. $F(s)$) for a stable network:
 1. The function should not have **any pole** in the right half of the s-plane.
 - The real part of poles should be negative
 2. The poles on the imaginary axis should be simple (not repeated).
 3. The degree of the numerator of $F(s)$ (i.e. m) should not exceed the degree of denominator (i.e. n) by more than unity → $m - n \leq 1$ (simple pole at infinity)



** recall: poles == roots of the denominator

Stability of Network Functions

Hurwitz Polynomial: Necessary conditions

- When it is not easy to get the roots of $D(s) = 0$ and hence the location of poles of $F(s) = \frac{N(s)}{D(s)}$, **the system stability is tested by checking if $D(s)$ is a Hurwitz polynomial**
- A Hurwitz polynomial $P(s) = a_n s^n + \dots + a_2 s^2 + a_1 s + a_0$ verifies the **necessary conditions**:
 - All the coefficients $a_0, a_1, a_2 \dots a_n$ are positive.
 - A polynomial may not have any missing terms between the highest and the lowest order unless all even or all odd terms are missing (i.e. necessary condition for having the roots of $P(s)$ have real parts which are zero or negative).

** note: if $P(s) = H(s)K(s)$ and $H(s), K(s)$ are Hurwitz polynomials the $P(s)$ is Hurwitz polynomial
- A polynomial is known as **strictly Hurwitz** if it does not have any **zero** in the right half of the s-plane or on the imaginary axis.

Stability of Network Functions

Hurwitz Polynomial: Necessary and **Sufficient** conditions

- After testing the necessary conditions (previous slide) continue with the **Test of Hurwitz**
 - Perform a continued fraction expansion of the **ratio of the odd to even parts** or **even to odd parts** of Hurwitz polynomial
 - ✓ **Step 1:** Arrange $P(s) = P_e(s) + P_o(s)$,
 - ✓ **Step 2:** then start with $\frac{P_e(s)}{P_o(s)}$ if the degree of $P(s)$ is even and with $\frac{P_o(s)}{P_e(s)}$ otherwise
 - ✓ **Step 3:** Get the quotient and the remainder of the division
 - ✓ **Step 4:** divide the divisor by obtained remainder and repeat steps 3 and 4 until getting a zero remainder
 - The continued fraction should give **all positive quotient terms** and **their number is equal to the degree of $P(s)$** .
 - If the continued fraction terminates abruptly (number of quotient terms is not equal to the degree of $P(s)$), the **last obtained non zero remainder $C(s)$** is a common factor between $P_e(s)$ and $P_o(s)$. If this term is Hurwitz then $P(s)$ is Hurwitz.
($P(s) = C(s)P_1(s)$)
 - If the ratio of the polynomial $P(s)$ and its derivative $P'(s)$ gives a continued fraction expansion with all positive coefficients then the polynomial $P(s)$ is Hurwitz → use this property for testing $P(s)$ when this later is purely even or odd polynomial

$$\begin{array}{r|l} P_e(s) & P_o(s) \\ \text{Divident} & \text{Divisor} \\ \hline & \text{Quotient} \\ \hline & \text{Remainder} \end{array}$$

Example 3

Test whether the following polynomials are Hurwitz

a. $s^5 + 3s^3 + 2s^2 + 3s + 1$

b. $s^3 + 6s^2 + 12s + 8$

c. $s^4 + s^3 + 2s^2 + 3s + 2$

d. $s^5 + 5s^3 + 4s$

e. $s^7 + 2s^6 + 2s^5 + s^4 + 4s^3 + 8s^2 + 8s + 4$

Specific Examples of Stable Network Function

■ All pass function:

- **All its poles** are located in the left half of s-plane only (real part of s (i. e. α) < 0)
 - **For each pole there is a zero** mirror image about the origin in the right half of the s-plane
- The function doesn't have any pure imaginary poles or zeros → strict Hurwitz polynomial
- An important property of all pass function is that its magnitude response is constant for all real frequencies (i.e. $|F(s)|$ is constant for all $s = j\omega$)

■ Minimum phase function:

- **All its poles and zeros** are located in the left half of s-plane or at the imaginary axis (real part of s (i. e. α) ≤ 0)

Decomposition of Stable Network Function into all-pass and minimal phase functions

- A stable network function $F(s)$ can be written as product of an all-pass function ($P(s)$) and a minimal phase function ($H(s)$)

$$F(s) = P(s)H(s)$$

1. Group the zeros located in the right half of the s-plane

$$\rightarrow F(s) = (s - a)(s - z_i)(s - \bar{z}_i) \cdot F_1(s)$$

With $F_1(s)$ doesn't contain zeros in the right half of s-plane (all its poles and zeros are in the left half)

2. Multiply and divide by $(s + a)(s + z_i)(s + \bar{z}_i)$ (i.e. the product of the mirror of the zeros $a, z_i, \bar{z}_i, \dots$ with respect to the origin)

$$\rightarrow F(s) = \frac{(s-a)(s-z_i)(s-\bar{z}_i)}{(s+a)(s+z_i)(s+\bar{z}_i)} \cdot (s + a)(s + z_i)(s + \bar{z}_i)F_1(s)$$

$$\rightarrow P(s) = \frac{(s-a)(s-z_i)(s-\bar{z}_i)}{(s+a)(s+z_i)(s+\bar{z}_i)} \text{ and } H(s) = (s + a)(s + z_i)(s + \bar{z}_i)F_1(s)$$

Example 4

Which of the network functions below is:

- a. an all-pass function
- b. A minimum phase function

$$F_1(s) = \frac{s^3 - 4s^2 + 14s - 20}{s^3 + 4s^2 + 14s + 20}$$

$$F_2(s) = \frac{s^3 + 2s^2 + s + 2}{s^3 + 4s^2 + 14s + 20}$$

$$F_3(s) = \frac{s^2 - s + 6}{s^3 + 4s^2 + 14s + 20}$$

Outline

- Introduction to Network Functions
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- **Derivation of Network Function from Partial Information**

Derivation of Network Function from its Real Part

- $U(\omega) = F_e(s)|_{s=j\omega}$ if $j\omega$ is in the region of convergence of $F_e(s)$ (i.e. region in the s-plane with no poles)
with $F_e(s)$ is the even part of $F(s)$ given by $F_e(s) = \frac{1}{2}(\mathbf{F(s)} + F(-s))$
- Express $F_e(s) = U\left(\frac{s}{j}\right)$
- Decompose $F_e(s)$ into partial fractions
$$F_e(s) = \frac{a_ms^m + a_{m-1}s^{m-1} + a_{m-2}s^{m-2} + \dots + a_1s + a_0}{b_ns^n + b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0} = \frac{A_i}{s-P_i} + \frac{B_i}{s-Q_i} + \frac{C_i}{s^2+\omega_i^2} + K$$
- We can obtain:
 - Terms with poles having a negative real part (i.e. $\frac{A_i}{s-P_i}$ with $\text{Re}(P_i) < 0$)
 - Terms with poles having a positive real part (i.e. $\frac{B_i}{s-Q_i}$ with $\text{Re}(Q_i) > 0$)
 - Terms with pure imaginary poles (real part is zero) (i.e. $\frac{C_i}{s^2+\omega_i^2}$)
 - $\lim_{s \rightarrow \infty} F_e(s) = K(\text{constant})$

$$\rightarrow F(s) = 2 \left(\sum_{\text{Re}(P_i) < 0} \frac{A_i}{s-P_i} \right) + \sum \frac{C_i}{s^2+\omega_i^2} + K$$

Proof

$$F_e(s) = \frac{1}{2}(F(s) + F(-s)) = \frac{1}{2}(P_n(s) + P_{PI}(s) + K + P_p(s) + P_{PI}(s) + K)$$

$$\frac{1}{2}(P_n(s) + 2P_{PI}(s) + 2K + P_p(s))$$

$$\frac{1}{2}P_n(s) + P_{PI}(s) + K + \frac{1}{2}P_p(s)$$

P_n : Poles with negative real part
 P_p : Poles with negative real part
 P_{PI} : Pure Imaginary Poles

From partial fraction decomposition : $F_e(s) = P_{en}(s) + P_{ep}(s) + K + P_{ePI}(s)$

By identification:

$$\frac{1}{2}P_n(s) = P_{en}(s)$$

$$P_{PI}(s) = P_{ePI}(s)$$

$$K = K$$

And finally **$F(s) = P_n(s) + P_{PI}(s) + K = 2P_{en}(s) + P_{ePI}(s) + K$**

Derivation of Network Function from its Imaginary Part

- $jV(\omega) = F_o(s)|_{s=j\omega}$ if $j\omega$ is in the region of convergence of $F_e(s)$ (i.e. region in the s-plane with no poles)
with $F_o(s)$ is the odd part of $F(s)$ given by $F_o(s) = \frac{1}{2}(F(s) - F(-s))$
- Express $F_o(s) = jV\left(\frac{s}{j}\right)$
- Decompose $F_o(s)$ into partial fractions
$$F_o(s) = \frac{a_ms^m + a_{m-1}s^{m-1} + a_{m-2}s^{m-2} + \dots + a_1s + a_0}{b_ns^n + b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0} = \frac{A_i}{s-P_i} + \frac{B_i}{s-Q_i} + \frac{C_i s}{s^2 + \omega_i^2} + K_\infty s$$
- We can obtain:
 - Terms with poles having a negative real part $\left(i.e. \frac{A_i}{s-P_i} \text{ with } \text{Re}(P_i) < 0\right)$
 - Terms with poles having a positive real part $\left(i.e. \frac{B_i}{s-Q_i} \text{ with } \text{Re}(P_i) > 0\right)$
 - Terms with pure imaginary poles (real part is zero) $\left(i.e. \frac{C_i s}{s^2 + \omega_i^2}\right)$
 - Pole at infinity : Term $K_\infty s$ with $K_\infty = \lim_{s \rightarrow \infty} \frac{F_o(s)}{s}$

$$\rightarrow F(s) = 2 \left(\sum_{\text{Re}(P_i) < 0} \frac{A_i}{s-P_i} \right) + \sum \frac{C_i s}{s^2 + \omega_i^2} + K_\infty s$$

Proof

$$\begin{aligned}
 F_o(s) &= \frac{1}{2} (F(s) - F(-s)) = \frac{1}{2} (\mathbf{P_n(s)} + \mathbf{P_{PI}(s)} + \mathbf{K_\infty s} - (P_p(s) - \mathbf{P_{PI}(s)} - \mathbf{K_\infty s})) \\
 &= \frac{1}{2} (P_n(s) + 2P_{PI}(s) + 2K_\infty s - P_p(s)) && P_n: \text{Poles with negative real part} \\
 &= \frac{1}{2} P_n(s) + P_{PI}(s) + K_\infty s - \frac{1}{2} P_p(s) = F_o(s) && P_p: \text{Poles with negative real part} \\
 &&& P_{PI}: \text{Pure Imaginary Poles}
 \end{aligned}$$

From partial fraction decomposition: $F_o(s) = P_{on}(s) + P_{op}(s) + P_{oPI}(s) + K_\infty s$

By identification:

$$\frac{1}{2} P_n(s) = P_{on}(s)$$

$$P_{PI}(s) = P_{oPI}(s)$$

$$K_\infty = K_\infty$$

And finally, $\mathbf{F(s) = P_n(s) + P_{PI}(s) + K_\infty s = 2P_{on}(s) + P_{oPI}(s) + K_\infty s}$

Derivation of Network Function from its Magnitude

- $A(\omega) = |F(s)| \big|_{s=j\omega}$

- $A^2\left(\frac{s}{j}\right) = |F(s)|^2 = F(s) \cdot F(-s)$

- *** $\overline{F(\bar{s})} = F(\bar{s}) = F(-s)$ reflection property at real frequencies)

- $A^2(s)$ can have:

- Poles having a negative real part → They belong to $F(s)$
- Poles having a positive real part → They belong to $F(-s)$
- Pure imaginary poles (real part is zero) → They belong to $F(s)$ and $F(-s)$ → These poles should appear with multiplicity 2 → attributed to $F(s)$ with multiplicity 1
- Pole at infinity belongs also to $F(s)$ and $F(-s)$ and hence should appear with multiplicity 2
- Zeros with real part $\text{Re}(Z_i) \neq 0$ → They belong either to $F(s)$ **or** to $F(-s)$ (each distribution of zeros among $F(s)$ and $F(-s)$ gives a possible solution for the problem)
- Pure imaginary zeros (real part is zero) → They belong to $F(s)$ **and** $F(-s)$ → These zeros appear with any **even multiplicity m** → attributed to $F(s)$ with multiplicity $m/2$

Example 5

Find the network function in each of the following

- a. Real part is known: $U(\omega) = \frac{\omega^2 + 2}{(\omega^2 + 1)(1 - \omega^2)}$
- b. Imaginary part is known: $V(\omega) = \frac{\omega^3}{(\omega^2 + 4)}$
- c. Magnitude is known: $A^2(\omega) = \frac{\omega^2(\omega^2 + 1)}{(\omega^2 - 1)^2(\omega^2 + 4)}$
- d. Magnitude is known: $A^2(\omega) = \frac{\omega^4}{(\omega^4 + 1)}$